

Camera Model Setup – Lecture Notes

1. What is a Camera Model?

A camera model explains how a 3D real-world point (X, Y, Z) is converted into a 2D image point (x, y). In simple words, the camera maps real-world coordinates into image pixel coordinates.

2. Pinhole Camera Model

The pinhole camera model is the simplest mathematical model of a camera. Light passes through a small hole and forms an image on the image plane.

Projection Equations:

$$x = f * (X / Z)$$

$$y = f * (Y / Z)$$

Where: f = focal length, Z = depth (distance from camera). If Z increases, the object appears smaller. If Z decreases, the object appears larger.

3. Camera Coordinate Systems

There are three coordinate systems: 1) World Coordinates (real-world position), 2) Camera Coordinates (relative to camera), 3) Image Coordinates (pixel position).

4. Extrinsic Parameters (R and T)

Extrinsic parameters describe the position and orientation of the camera in the world. They include Rotation (R) and Translation (T).

$$X_c = R * X_w + T$$

5. Intrinsic Parameters (Camera Matrix K)

Intrinsic parameters describe the internal properties of the camera such as focal length and principal point.

f_x	0	c_x
0	f_y	c_y
0	0	1

6. Numerical Example

Suppose $f = 50$ mm and the object point is $(X=2\text{m}, Y=1\text{m}, Z=5\text{m})$.

$$x = 50 * (2/5) = 20$$

$$y = 50 * (1/5) = 10$$

Therefore, the image point is $(20, 10)$.

7. Complete Camera Model Equation

$$x = K [R \mid T] X$$